

Tracking Error as an Observation for Behavior Cloning on Low-Cost Arms

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Abstract: When a robot records commanded and measured joint positions, their difference provides an observation of actuator response that can be reconstructed without an additional sensor or a fitted force estimator. We study this tracking error as an additional input to behavior cloning on a low-cost SO-101 arm. In simulation, placement success increases from 6.2% with position observations to 21.8% with tracking error across five training seeds, but position history reaches 20.3% across three seeds, leaving the specific benefit of tracking error unresolved. On hardware, policies trained on the same 50 demonstrations achieve 9/20 and 14/20 placements with tracking error against 5/20 and 1/20 for their position-only counterparts in two completed evaluations. These results support tracking error as a useful candidate observation while exposing substantial variation across training seeds and limits to its interpretation as contact feedback, and a study of archived teleoperation data shows why large tracking error alone is an unreliable grasp-success label.

Keywords: Behavior Cloning, Proprioception, Low-Cost Manipulation

1 Introduction

A position command describes an intended motion, and during manipulation the gap between that command and the position actually reached can persist for many steps—because a finger has met an object, because a joint is carrying a load, or simply because the actuator has not caught up. A policy given only the current joint position must infer all of this from its remaining observations, whereas supplying the previous command makes the discrepancy directly available.

This is an attractive observation for inexpensive robots because it requires neither an added sensor nor a force-estimation model, and wherever recorded commands correspond to the goals applied to the robot at a known cadence it can be reconstructed from demonstrations that already exist. Convenience is not value, though: tracking error mixes contact with motion lag and actuator effects, and richer temporal inputs may carry the same useful information. The practical question is whether exposing this discrepancy helps an imitation policy, and what evidence would justify attributing any benefit to contact.

We investigate that question through a simulation comparison, which lets us set explicit tracking error against raw action history, position history and a compensated residual, and a small hardware study, which asks whether the simplest of those features transfers to human demonstrations on one SO-101. The two completed hardware comparisons favour tracking error. The experiments motivate retaining command history in policy interfaces, and they fall well short of a generally reliable force-aware controller.

2 Related work

Sensorless force estimation has a longer history than its use in learned manipulation. Hwang et al. [1] develop a virtual torque sensor for RC servos using dynamic system identification. More recently,

FACTR 2 [2] learns external-torque estimates from free-motion data and uses force-informed re-sampling to improve policy learning. We study a simpler input, the uncalibrated command–position difference itself, which requires no fitted estimator and makes correspondingly weaker claims about physical force.

Wong et al. [3] show that leader–follower discrepancies in teleoperation can carry interaction cues: removing them from ACT’s action labels degrades performance in force-critical phases, while torque proxies restore contact-aware behavior. Our intervention instead adds the discrepancy to the policy observation, which makes this an observation-design study rather than a new sensorless-force principle. The distinction matters statistically as well, since adding the feature adds access to a previous command, and history that helps under partial observability can equally support shortcuts that predict previous actions instead of the actions the task needs [4]. We therefore compare several temporal observations in simulation rather than reading every improvement over a position-only baseline as evidence for force estimation.

3 Observation and experimental design

Let q_t be the measured joint position and g_{t-1} the preceding applied position goal, expressed in the same coordinate system. We define

$$\delta_t = g_{t-1} - q_t, \quad o_t^{\text{base}} = [I_t, \tilde{q}_t], \quad o_t^\delta = [I_t, \tilde{q}_t, \delta_t/c], \quad (1)$$

where I_t is the camera observation and $\tilde{q}_t$ uses training-set normalization. The scale $c = 8$ is fixed in each dataset’s joint-coordinate units and is not a conversion to torque. Under approximately static contact, tracking error may reflect load on a position servo, but during motion it also contains following error, so we use it as an observation without assigning every nonzero value to external contact.

Simulation. We train ACT policies [5] on 280 scripted pick-and-place demonstrations with randomized object and scene properties. All designs use the same data and training budget: 12,000 steps with 96-pixel front and gripper-camera images. Each trained policy is evaluated on 100 episodes with shared scene seeds across designs, five training seeds being available for the original designs and three for the position-history control. The learning comparison uses placement success with the crush threshold disabled, so it tests grasping and transport rather than regulation within a safe force band.

Hardware. We collect 50 teleoperated demonstrations of placing an adapter inside a tape roll using one SO-101 and one overhead camera. Both policies use the same 18,549 training frames after removal of stationary prefixes, 96-pixel images, and 6,000 training steps. The proprioceptive input has six dimensions for the base and twelve for tracking error; all other architecture and optimization settings are shared. Three seeds are trained, one checkpoint per observation design and seed.

Evaluation begins after replaying 115 commands from a fixed demonstration, which places the arm near the grasp, after which each policy controls up to 400 steps. ACT predicts and executes 30-command chunks, so observations inform the next chunk rather than each command within it. A relative-target limiter is shared by both policies, and the tracking-error input uses the command returned *after* limiting. Trials are paired by training seed and physical reset, with base-first and delta-first order alternating across pairs, and an operator records whether the adapter is placed inside the roll. We plan 20 pairs per seed and retain failures, including outcomes from completed rollouts whose shutdown reports an error. Appendix A gives implementation and interruption details.

4 Results

Temporal information helps in simulation; its best representation is unresolved. Mean placement success is 6.2% for the position-only base, 13.8% with the previous action appended, and 21.8% with explicit tracking error. A compensated residual reaches 26.6%, while position history alone reaches 20.3% (Fig. 1). Because the residual subtracts a fitted free-motion term from tracking

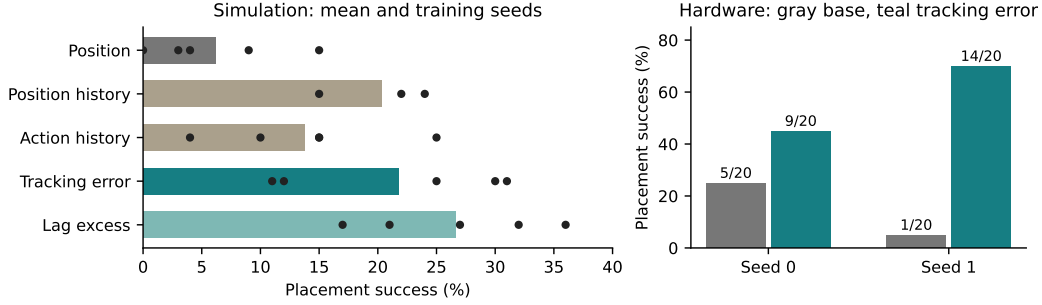


Figure 1: Tracking error improves on position-only observations in simulation and in two completed hardware comparisons. Left: bars are means; dots are individual training seeds (five per design, three for position history). Right: hardware success counts for the two completed evaluations. The full evaluation record, including an interrupted third comparison, appears in Appendix A.

Table 1: Hardware placements and paired outcomes for the two completed evaluations. Discordances list tracking-error-only / base-only successes. See Appendix A for the interrupted evaluation.

Seed	Evaluation	Base	Tracking error	Discordances	<i>p</i>
0	Complete	5/20	9/20	6 / 2	0.2891
1	Complete	1/20	14/20	14 / 1	0.0010

error and uses two previous commands, comparing it against one-command history changes both the information available and its representation. More importantly, position history performs comparably to tracking error without observing commands at all, so the grid supports supplying temporal information without resolving whether explicit tracking error is preferable to a suitable history. Predicting tracking error as an auxiliary training target gives 6.6%: this particular auxiliary objective does not recover the benefit of supplying it as input.

Hardware improvements depend on the trained policy. Table 1 reports each comparison separately, and tracking error improves placement by 20 percentage points for seed 0 and 65 points for seed 1. The exact two-sided McNemar tests compare discordant pairs within each seed: six versus two favour tracking error in seed 0, and fourteen versus one in seed 1, so the first comparison is inconclusive while the second shows a substantial difference between these two checkpoints. These tests quantify trial variation conditional on a trained policy, not variation over training seeds.

A third evaluation was interrupted by recurring hardware faults; its partial results and missing outcomes are reported in Appendix A.

A useful input is not necessarily an outcome label. We also test a preregistered prediction that larger tracking error at detected grasp onset accompanies successful grasps in archived teleoperation data. Sixteen datasets with 546 episodes pass the inclusion gates, of which only two meet the predicted positive effect-size threshold; the pooled within-dataset effect is negative ($d = -0.39$), scored against a gripper-retention proxy rather than manually verified placements. The result rejects this proposed labeling rule and illustrates why a discrepancy can indicate resistance without identifying a successful grasp (Appendix E).

5 Discussion and conclusion

Tracking error appeals for a practical reason: it exposes actuator response using signals the command interface already carries. The simulation comparison and the two completed hardware comparisons show that it can help behavior cloning at a small demonstration budget, but nothing here establishes that the policy learns force regulation—position history is competitive in simulation, no history

control has been evaluated on hardware, and the hardware task never requires respecting a measured force threshold.

Variation across seeds is central to the result: a large advantage for one checkpoint pair coexists with a smaller advantage for another. Covering one physical arm, one task and a fixed demonstration-assisted start, the study cannot speak to end-to-end task reliability, transfer across objects or robots, or independence from the command limiter. Observing a contact-related signal does not itself enforce acceptable loading.

Tracking error is therefore a useful candidate for observation design, with an empirical benefit that should be tested against temporal baselines and across training seeds. When it adds information beyond history, and whether a policy can use that information to regulate contact, are the two questions we would put to the next round of experiments.

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A Hardware protocol and complete evaluation record

Table 2: Full hardware evaluation record. The interrupted seed has six recorded pairs, with 0/6 placements for each policy; its planned budget was 20 pairs. No confirmatory test is assigned to the incomplete comparison.

Seed	Evaluation	Base	Tracking error	Discordances	p
0	Complete	5/20	9/20	6 / 2	0.2891
1	Complete	1/20	14/20	14 / 1	0.0010
2	Interrupted	0/6	0/6	0 / 0	—

Training. The recording contains 21,476 frames from 50 demonstrations at 30 Hz. Position and action statistics are computed on the recording; the training index then removes stationary prefixes, retaining 18,549 frames. Both policies use the same index and normalization, and recorded load channels are excluded from policy inputs. The real-data coordinates follow the recording interface; in particular, gripper position is not a calibrated force or a common angular coordinate shared with the arm joints.

The ACT configuration uses a ResNet-18 image backbone, model width 256, four attention heads, feedforward width 1024, two encoder layers and one decoder layer. The action chunk and execution horizon are both 30 steps, with no temporal ensembling. Optimization uses AdamW at learning rate 10^{-4} , weight decay 10^{-4} , batch size 8 and gradient clipping at 10. All reported checkpoints are the final checkpoint at 6,000 steps. The observation projection necessarily changes width when six features are added; the remaining architecture and training settings are shared.

Execution and command semantics. Each rollout homes the arm to the mean demonstration start, replays the first 115 actions from demonstration 0, and hands control to the policy, so the reported task excludes autonomous approach from an arbitrary initial state. Each requested command passes through a relative-target limit of 8 in the interface’s joint units—a target-displacement bound rather than a force or velocity guarantee—and during both replay and policy execution the returned limited command supplies the next tracking-error calculation. The servo’s native goal quantization sits below this software interface. State and images are read at each control step, but ACT consults them only when its 30-step action queue is empty.

An offline replay check reproduces the training and inference state vectors on all 21,476 recorded frames for the base, tracking error, action history, position history and compensated residual designs. In seed 0, the saved tracking-error observations after the first rollout step also reproduce from logged applied commands, which checks software consistency rather than physical command-to-encoder latency. Collection-time command limiting has not been established from the available metadata, so archived action labels should be read as recorded goals, which need not equal the goals applied to a servo if collection used an additional limiter.

Pairing and inference. Hardware resets are manual, and within each seed consecutive trials form a pair with AB/BA order alternating between pairs. The operator knows the policy identity, so outcomes are not blinded. Seeds 1 and 2 were scheduled after observing seed 0, with their training recipe and evaluation budget fixed before their evaluations, and no policy was selected by hardware performance. The primary paired test is exact, two-sided McNemar with an improvement criterion of positive effect and $p < 0.05$; for the completed seeds, paired percentile bootstrap intervals (20,000 pair resamples, random seed 0) are $[-5, +45]$ and $[+40, +90]$ percentage points, describing trial uncertainty for fixed checkpoints rather than confidence intervals over retraining. The equal-weight mean of the three observed seed differences is 28.3 points, but the third difference uses only six pairs and is censored by hardware failure, so we do not read that mean as a balanced estimate of method performance.

Interruptions and missing data. Seed 0 has 40 outcomes and 40 trajectories; seed 1 has 40 outcomes and 39. Trials 35 (tracking error) and 37 (base) completed their control horizons but reported a gripper overload while disabling torque, and both are operator-labeled failures. Trial 35’s out-

come was recovered from the operator after the process stopped, but its trajectory was lost because the earlier cleanup path saved data only after disconnecting. Saving was moved before disconnect, without changing policy observations or commanded actions, and trial 37’s trajectory was retained. Evaluation resumed with the original pair identifiers and order.

Seed 2 has twelve labeled rollouts, all failures, forming six complete pairs, with shutdown overloads recorded on trials 8, 10 and 12. Trial 13 also completed its control horizon and has a trajectory and shutdown-failure record, but its outcome was not entered, and the remaining planned trials were never evaluated. We preserve the unrecorded outcome as missing rather than infer it from joint motion. We have established neither a diagnosis of the recurring overload nor random missingness, and this interrupted evaluation is part of the evidence against claiming broad hardware reliability.

B Exploratory trajectory analysis

The seed-0 tracking-error successes follow similar jaw trajectories and settle near a common final position, while failures show more varied jaw motion. Over the last 100 steps, median summed joint-coordinate travel—a descriptive mixed-coordinate sum rather than a physical path length—is 14.3 for successes and 93.6 for failures, and the median fractions of steps affected by command limiting are 26% and 5.75% respectively. More limiting in successful trials could simply follow from progressing further through the task. The traces were grouped by observed outcome after evaluation and contain no video, object pose or ground-truth force, so they cannot identify a missed grasp, a slip or a placement error, and the endpoint separation is not a validated success detector.

C Measurement context

Tracking error is not introduced here as a new torque sensor. A separate static experiment on one shoulder-lift STS3215 uses twelve hanging masses at five poses (60 measurements). Across those measurements, the fitted relationship between the load register and tracking error is $\text{Present_Load} = -4.70\delta + 0.1$, with $R^2 = 0.976$. Its relation to applied mass is far less stable, with per-pose R^2 for tracking error ranging from 0.085 to 0.910, so the tight load-register association does not establish an equally accurate physical-load measurement.

In the 50 teleoperation demonstrations, the gripper load register reaches 500 in every episode and remains there in 16.5% of frames, while tracking error continues to vary across those same frames. The inspected follower configuration sets a reduced gripper torque limit, consistent with that observed plateau. This motivates inspecting the command–position discrepancy when telemetry saturates, but it does not calibrate the remaining discrepancy in newtons: no jaw load-cell calibration is available, and simulator force conversions such as the fitted 2.00 N/count slope and 4.3 N held-out RMSE over 4–78 N describe the simulated plant only.

D Additional simulation and hardware comparisons

Simulation recordings use integer encoder counts at 15 Hz, obtained by subsampling the 30 Hz controller. The scripted teacher regulates the jaw using tracking error, and the successful demonstration set includes randomized grip depth and recovery perturbations. These collection choices are shared across observation designs, of which the position-history control observes $[q_t, q_{t-8}]$ and one-command history observes $[q_t, g_{t-1}]$. Neither comparison makes the simulation and hardware tasks identical: camera views, demonstrations and recording cadence all differ between the two settings.

The original simulation study includes an auxiliary tracking-error target and a latched seat token in addition to the designs in Fig. 1, with mean placement rates of 6.6% and 21.8% across five seeds. The compensated residual is $e_t = \delta_t - \hat{K}(g_{t-1} - g_{t-2})$, where each joint’s coefficient is fitted once on presumed free-motion demonstration frames. A two-command history control reaches 14.7% over three seeds; comparing it with the residual holds available raw information fixed but does not match input width. Under the original study’s preregistered practical resolution rule, $15\sqrt{3/n}$ percentage

points for n training seeds, neither this comparison nor the residual–position-history comparison resolves at three seeds, and the stronger feature-specific interpretation remains open.

An earlier hardware comparison, run before the raw-error study, evaluated the compensated residual: the base placed 7/10 objects and the residual 0/10. The coefficients fitted from these real demonstrations differ markedly from the simulation fits, and a free-motion labeling heuristic includes contact frames. These are plausible sources of model mismatch rather than a causal diagnosis, since coefficients above one are not by themselves proof that a regression against one command increment is invalid, and the earlier runner also fed back requested rather than post-limit commands. That comparison differs in preprocessing and execution from the raw-error study and is reported separately; the later improvement does not isolate the compensation term as the cause of failure.

The simulation task admits a crush threshold, but the learning grid does not activate it. A post-hoc screen of existing checkpoints at 120 and 60 N produces heavy crushing and near-floor success, failing the predicted advantage of the compensated signal. A separate runtime guard reduces crush events from 85 to 3 across 180 crush-tier episodes, while eliminating all 12 unguarded placements across its broader 270-episode comparison. We do not present either result as successful learned force regulation or deploy that guard in the hardware comparisons.

Larger-budget experiments do not establish a scaling advantage. An earlier 600-demonstration comparison of raw tracking error and one-command history reaches 57% and 60%, respectively, over two seeds at 224 pixels and 50,000 steps. A later 600-demonstration screen saved at 26,000 steps contains three base and three residual evaluations, each with 0/100 placements. These floor results are reported separately from the 280-demonstration grid; they support neither a positive scaling claim nor equivalence between observations.

E Archived teleoperation study

The preregistered corpus prediction was that tracking error at a detected seat event would be larger in proxy-successful episodes, with Cohen’s $d > 0.5$ in at least ten of twenty targeted datasets. Sixteen datasets and 546 episodes pass the frozen gates, including 283 proxy successes under a success proxy of continued gripper closure through transport. Only two datasets exceed the positive threshold, and the pooled within-dataset effect is $d = -0.39$ —a failure of the proposed direction and labeling rule, though not proof that no function of command and position history can predict outcome.

Empty closure against a mechanical stop and closure on an object can both produce a seat event, and a large command discrepancy can equally reflect an operator continuing to close against resistance. These ambiguities explain why a contact-related feature need not be a success label, although the corpus does not isolate their causal contributions. Recorded leader commands may differ from applied follower goals and the timing is only approximately aligned, but an offset sweep from zero to five recorded frames leaves the qualitative conclusion unchanged, with median per-dataset effect-size shift 0.12.